

The barrier's cost is predicted from the asynchronous run alone

Asynchronous run



record every simulation's duration D_i
and the rate T_{async} ; predict $T_{\text{sync}} = W / \mathbb{E}[\max_{i \leq W} D_i]$ by resampling the D_i

Barrierized twin: same sampler + barrier

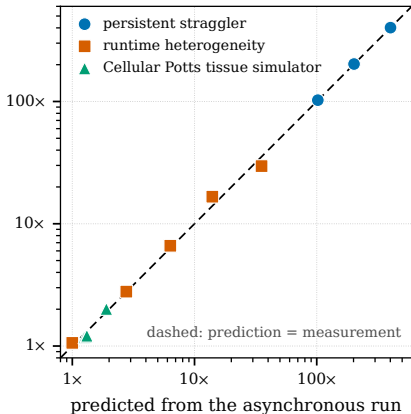


measure its rate T_{sync} directly; nothing else in the sampler changes

predicted
→

measured
→

measured with the barrierized twin



Predicted against measured barrier cost, 1.2× to 400×: within 1% under a persistent straggler, 20% under lognormal runtime heterogeneity, 10% on a Cellular Potts tissue simulator.